Saved factor bases: exact points, provenance and lookup caches

2026-10-16

The request is to preserve the exact factor bases used in the completed four-family evaluation, including their ordering and derivation, and make them reusable through a lookup cache. This additive historical import covers **all 384 cases and 52,876 canonical representative rows across those cases**. Points shared between bases are counted in each base's record. Each representative also has its negative in the cache. These are the historical 16-, 20- and 24-bit fields; the importer is pinned to that archive. No research campaign or performance benchmark was run, and no point logarithm was recovered.

What was already saved

PR #1898 preserved the full ordered `base` array in every `results/generic.json` record inside the frozen archive. It also preserved the generation program, protocol, seeds, curve parameters and family-specific metadata. The points were obtained by lifting coordinates onto $y^2 = x^3 - 3x + b$; they were **not generated by solving their logarithms**. The homogeneous relation study did not produce a table of point logarithms. Its archived `m` field counts base columns; the decomposition arity was three.

Those records were embedded in the larger experiment output. The new native exporter makes each base independently readable and builds a persistent signed-membership cache. The catalog binds each case to its full FB1 digest, trace digest, cache digest and order root. All original evidence remains unchanged.

The 48 historical curve models are registered in the existing ICV1 registry and native alias map. Their identities are computed by the existing Rust `curve_id::prime` implementation from the frozen parameters and recorded orders. The native source records feed the existing metadata assembler; no new point counting or Python curve arithmetic is used for this import. Existing registry records are preserved exactly, and the exporter refuses unregistered curve identities.

What each exported record contains

Record	Meaning
Source	Frozen archive member, exact SHA-256, array index, producer and protocol hashes
Curve	Existing native ICV1 identity, explicit equation and subgroup order
FB1	Existing `ecbench.factor_base/v1` identity convention and native group keys; full digest retained
Ordered representatives	Exact archived `(x,y)` and packed value at every original matrix column
Derivation	Family recipe, global seed, RNG seed inputs and recorded `base_extra`
Order root	Domain-separated SHA-256 Merkle tree binding curve, full FB1 digest, column and point

Record	Meaning
Cache	Both signs mapped to the original zero-based column and coefficient sign

The reference generator in the native curve wrapper is the first archived representative. This is a validation choice, not a historical generator claim or a known-scalar label. The cached sign is +1 or −1 relative to the archived representative; it is not a discrete logarithm.

The Merkle root commits to the exact ordered list. It does not prove how the points were generated. **Complete rejected-candidate events and the historical pre-shuffle acceptance order were not recorded.** The trace says so explicitly. The frozen producer is the authority for the recipe; its CPython RNG is not rerun or relabelled as a new native implementation.

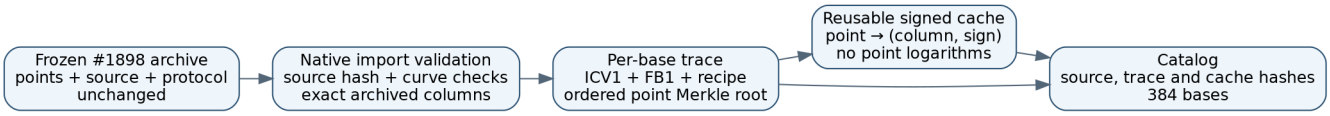


Figure 1: Frozen source to independently verifiable records and reusable cache.

Reuse and verification

Build the native tool, extract the existing frozen archive into a new directory, then export into a directory that does not exist yet:

```
cargo build --release --bin factor_base_trace --bin ecbench
fb_study_dir=research/factor_base_evaluation_20261015
unzip "$fb_study_dir/artifacts/factor_base_evaluation.zip" \
  -d /tmp/fb-history
target/release/factor_base_trace export \
  /tmp/fb-history/results/generic.json /tmp/fb-records
target/release/factor_base_trace verify \
  /tmp/fb-history/results/generic.json /tmp/fb-records
```

The exporter refuses a source whose bytes do not match the frozen SHA-256 recorded in the catalog and exporter. It refuses overwriting an existing export directory. Verify first against the frozen source when obtaining records from another location: self-consistent hashes alone do not authenticate a replacement dataset.

Each catalog entry's `name` identifies `NAME.json` and `NAME.cache`. Look up one or several points after loading and checking the cache once:

```
target/release/factor_base_trace lookup \
  /tmp/fb-records/000-FB1h….json /tmp/fb-records/000-FB1h….cache \
  X Y X2 Y2
```

Each output line is `[column,sign]` or `null` for an on-curve nonmember. Off-curve coordinates and cache corruption are errors. The actual filename is in the catalog; the ellipsis above is illustrative. The cache stores sorted native point keys, so each query uses binary search after loading and integrity checking. Loading remains proportional to the base size. This is an inventory and membership tool; it does not replace the decomposition oracle or provide a measured end-to-end speedup. Existing `ic.pipeline` still uses constructor specifications, as documented in the factor-base protocol.

Validation and storage status

All 384 records were exported from the pinned source and verified against it. Verification checks curve membership, `[r]P = 0`, canonical signs, uniqueness, exact point order, native identities, every catalog

field, cache bytes and lookup of **both signs of all 52,876 representatives**. Unit tests exercise signed column recovery, corruption, invalid/duplicate points and the distinction between set identity and order identity. These are artifact-validation results, not new research measurements; historical timing claims retain their original Sage/C provenance. No cross-method performance ranking is introduced.

The complete trace/cache bundle is saved in the durable store at `s3://crypto-autoresearcher/store`, under the indexed name `factor-base-traces/historical-20261015/registered-20261016.zip`. The store index records its full SHA-256 and 2,462,662-byte size. Upload succeeded and `store verify --remote` reported `cache ok, remote ok`. The initial attempt failed with `NoCredentials`; the authenticated retry passed. No credentials are part of the artifacts. The catalog and exporter are reviewable in Git; the original point payload also remains in the frozen Git archive and reproduces every trace/cache byte. The initial export remains indexed separately. Its points, ordering, derivation fields and cache bytes are identical; only the curve-registration status metadata changed before publication. Both objects are immutable.

With store access configured, retrieve the existing bundle without regenerating any points:

```
target/release/ecbench store pull \
  factor-base-traces/historical-20261015/registered-20261016.zip \
  --out /tmp/fb-traces.zip
unzip /tmp/fb-traces.zip -d /tmp/fb-records
```

The next experiment remains the separate, unrun draft `XP-20261016-prime-base-oracle-tuning` in PR #1963. This import neither funds nor leases that card. Future measured comparisons must follow current AGENTS.md pricing bands symmetrically and may count verified engineering improvements as successes without requiring a win over rho.